

Instructions

There are **24 total points**. When asked to provide your answer within a figure or table, be careful to not exceed box boundaries. Bubbles must be filled out completely: ● is correct, ☑ ☐ are incorrect. All answers must be given within the provided circles, answer boxes, figures or tables.

1. [1 point]: Write your full name in the box to acknowledge the instructions.

Machine Learning Pipeline

2. [3 points]: One of the challenges with training a machine learning model is overfitting—whereby a model performs well on the training data but poorly on the test data. Assume a “standard” traffic classification problem, like service identification or QoE determination. Give an example of a feature from network traffic which, if included in a model, could result in overfitting to the training data.

(Answer inside the box)

Supervised Learning

3. [2 points]: Random forest models can perform classification using a combination of feature types as input (e.g., categorical, numerical).

☐ Yes ☐ No

4. [2 points]: Deep learning models that operate on raw traffic input (like nPrint) always outperform simpler models like k-nearest neighbors and linear regression models that take semantic features as input.

☐ Yes ☐ No

5. [3 points]: Why or why not?

(Answer inside the box)

Unsupervised Learning

6. [2 points]: Which of the following are possible applications of unsupervised learning? (Select all that apply.)

- ☐ Detecting unusual traffic volumes or destinations
- ☐ Identifying changes that could indicate a network failure
- ☐ Identifying the application that generated a traffic trace.
- ☐ Clustering traffic traces into groups with similar behavior (e.g., to identify a botnet).

Initials: _____

7. [2 points]: Can you ever evaluate an unsupervised learning model using precision and recall?

☐ Yes ☐ No

8. [3 points]: If so, explain how. If not, why not?

(Answer inside the box)

9. [3 points]: Describe (with illustration, if you like) an example dataset where density-based clustering may produce more “meaningful” clusters than k-means clustering.

(Answer inside the box)

Looking Ahead

10. [1 point]: Transformers and large language models have shown promise in learning long-term temporal and semantic relationships across multiple traffic flows (e.g., a TCP connection that follows a DNS lookup, etc.). Can you think of a classification problem where an nPrint and simple classifier would fail but such a classifier model might perform better? (I know...I said I wouldn't ask, but it's only one point!)

(Answer inside the box)

Feedback

11. [2 points]: 1. Your favorite topic or activity in this course. 2. One topic you'd like to see covered that was not covered.

(Answer inside the box)

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